

TITLE:

PHQ/GAD score distribution analysis

Dataset Used:

anxiety_depression_data.csv

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DATA ANALYTICS INTERN

SUBMITTED TO:

ZENO TALENT

PHQ/GAD Score Distribution Analysis

PHQ (Patient Health Questionnaire) and GAD (Generalized Anxiety Disorder) scores are commonly used in mental health datasets to measure **depression severity** and **anxiety severity**.

Distribution analysis helps identify how these scores vary across different demographic groups such as **gender** and **age**.

Dataset Introduction

The dataset used for this project is **anxiety_depression_data.csv**, which contains mental health assessment information related to anxiety and depression disorders. The dataset includes demographic features such as:

- Age
- Gender
- Anxiety Scores (GAD)
- Depression Scores (PHQ)

The main objective of this analysis is to study how anxiety and depression scores are distributed across different genders and age groups. Statistical visualizations such as **box plots** and **violin plots** help identify:

- Median scores
- Score spread and variability
- Outliers
- Distribution density
- Age and gender-based mental health trends

SCORE DISTRIBUTION ANALYSIS:

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```
[7]: import pandas as pd
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv("/content/anxiety_depression_data.csv")

# Create age bands
bins = [0, 18, 30, 45, 60, 100]
labels = ['0-18',
          '19-30',
          '31-45',
          '46-60',
          '60+']

df["Age_Band"] = pd.cut(
    df["Age"],
    bins=bins,
    labels=labels,
    include_lowest=True
)

# -----
# Anxiety Score Box Plot by Gender
# -----

plt.figure(figsize=(8,5))

gender_groups = [
    df[df["Gender"] == g][["Anxiety_Score"]].dropna()
]
```

Variables Terminal

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```
[8]: # -----
# Anxiety Score Box Plot by Gender
# -----

plt.figure(figsize=(8,5))

gender_groups = [
    df[df["Gender"] == g][["Anxiety_Score"]].dropna()
    for g in df["Gender"].unique()
]

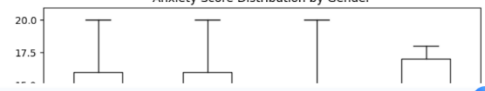
plt.boxplot(
    gender_groups,
    labels=df["Gender"].unique()
)

plt.title("Anxiety Score Distribution by Gender")
plt.xlabel("Gender")
plt.ylabel("Anxiety Score")

plt.show()
```

... /tmp/spykernel_396/3971055186.py:12: MatplotlibDeprecationWarning: The 'labels' parameter of boxplot() has been renamed 'tick_labels' since Matplotlib 3.9; support for the old name will be dropped in 3.10.

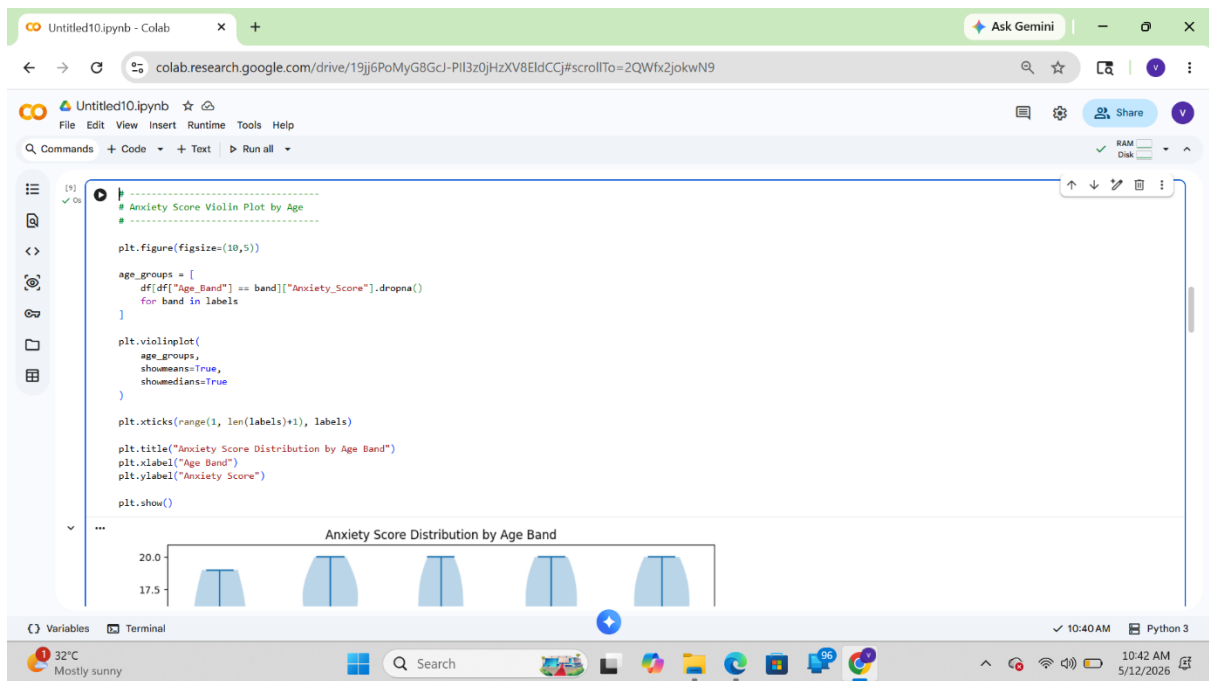
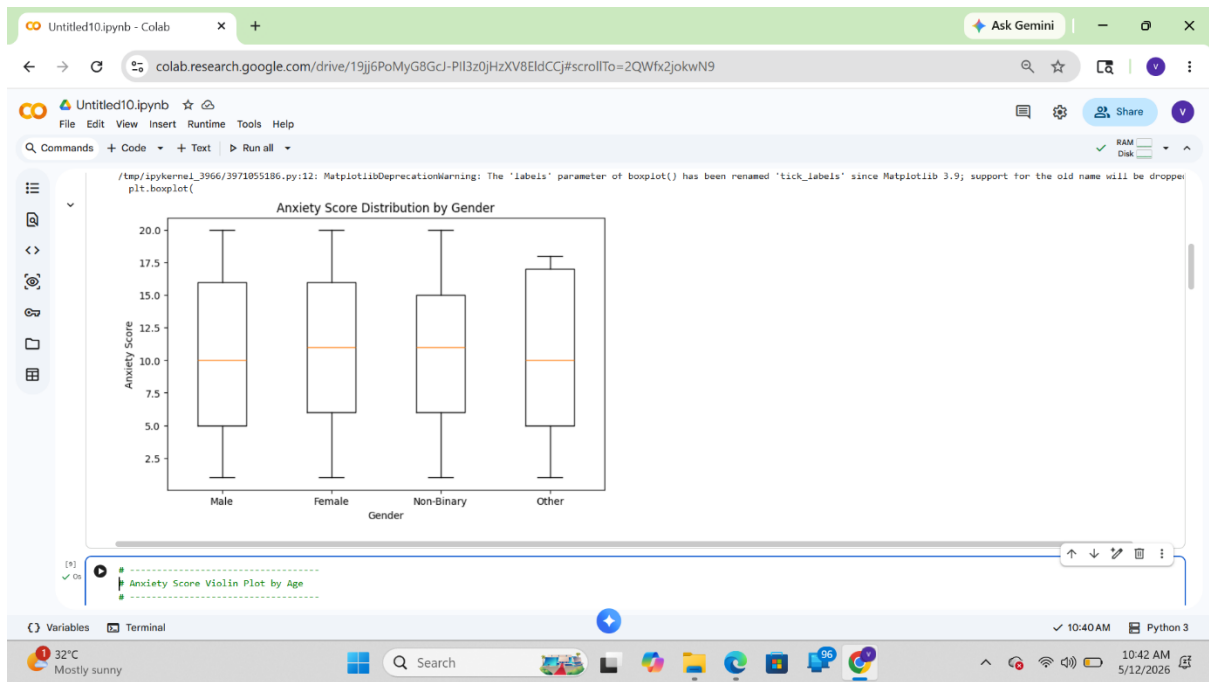
Anxiety Score Distribution by Gender

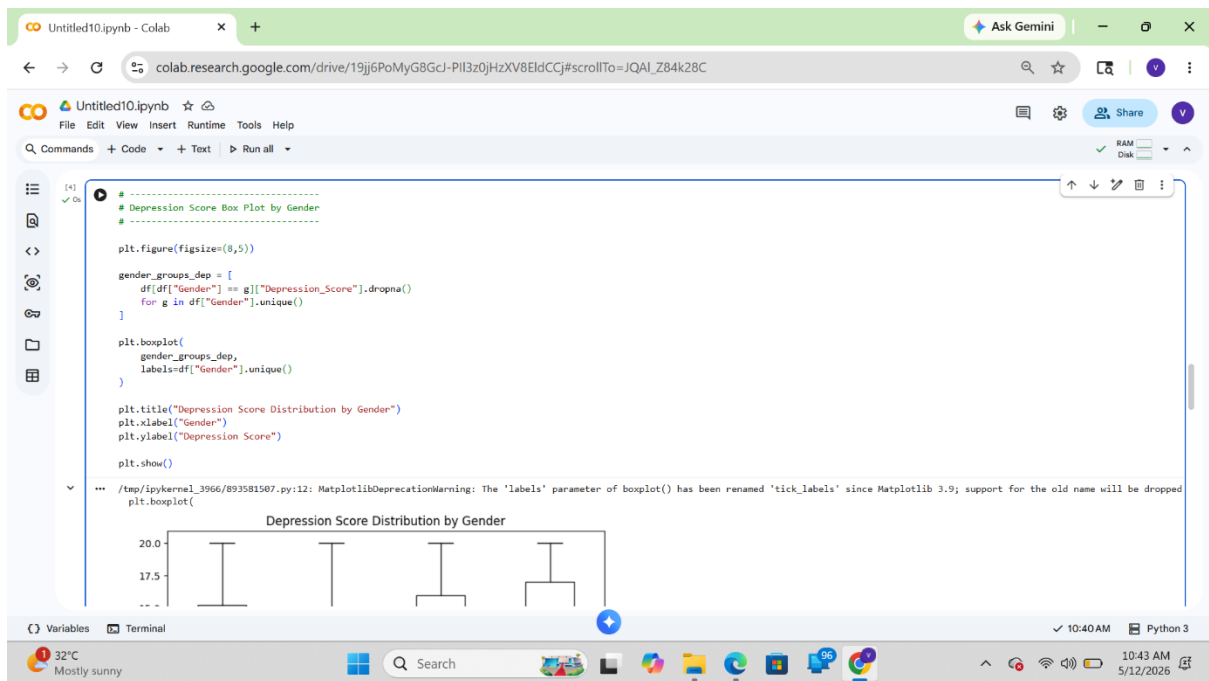
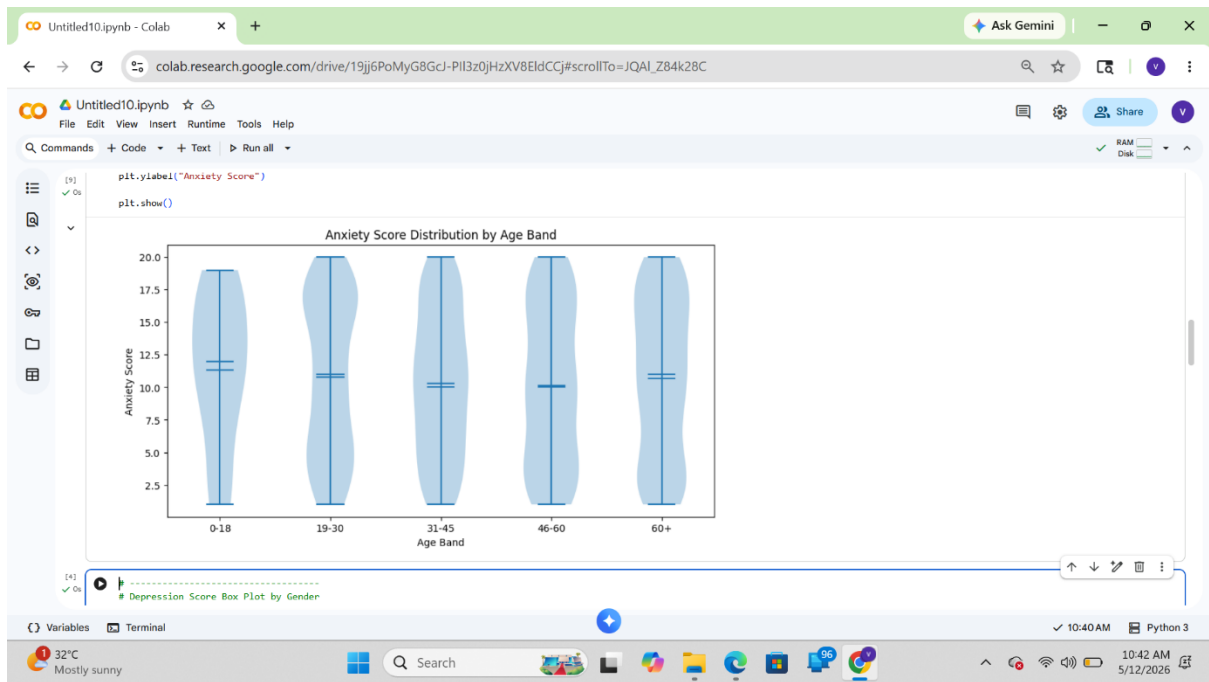


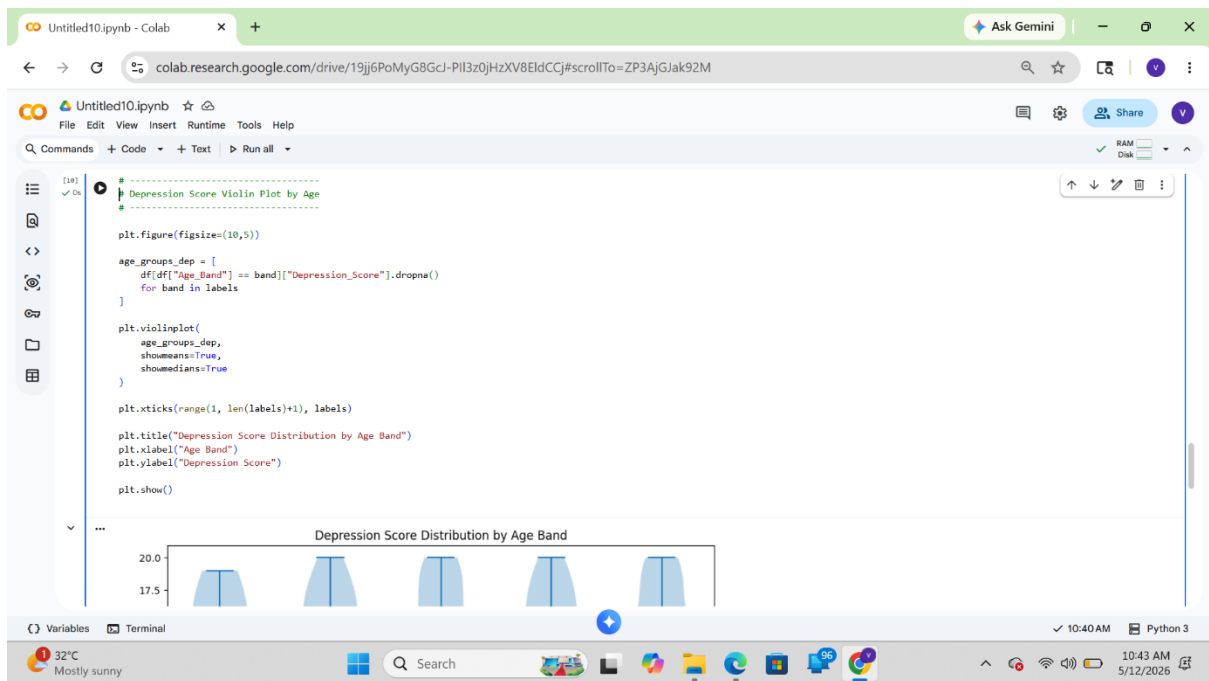
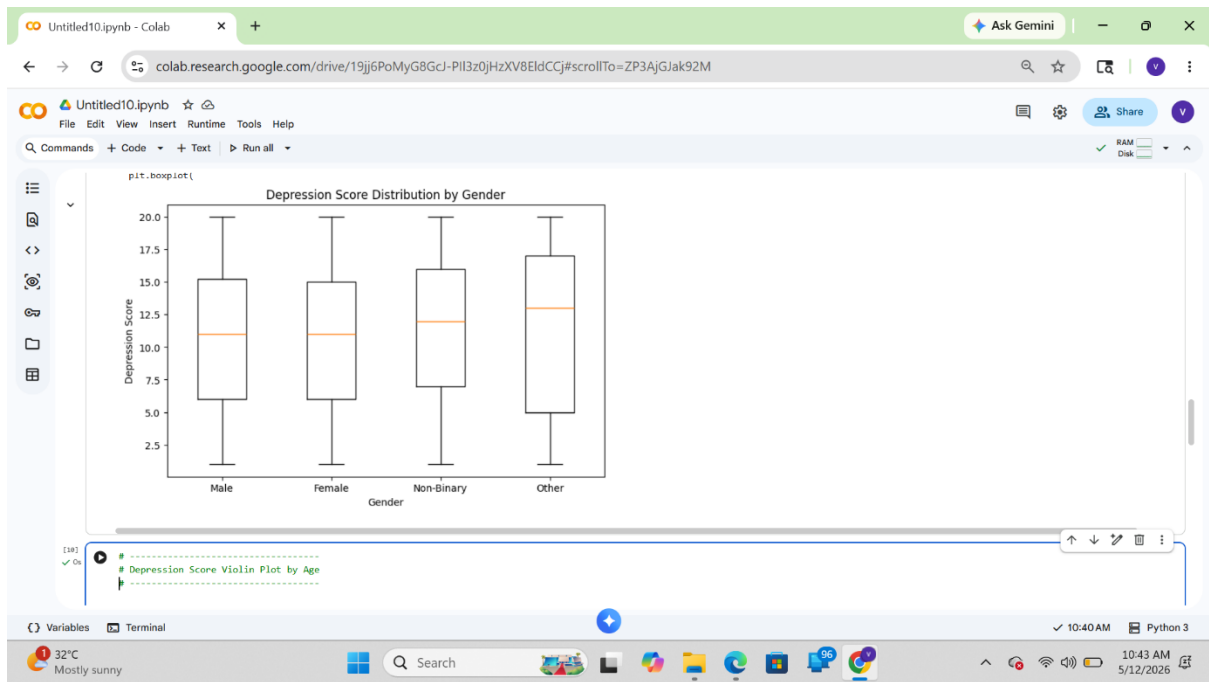
Variables Terminal

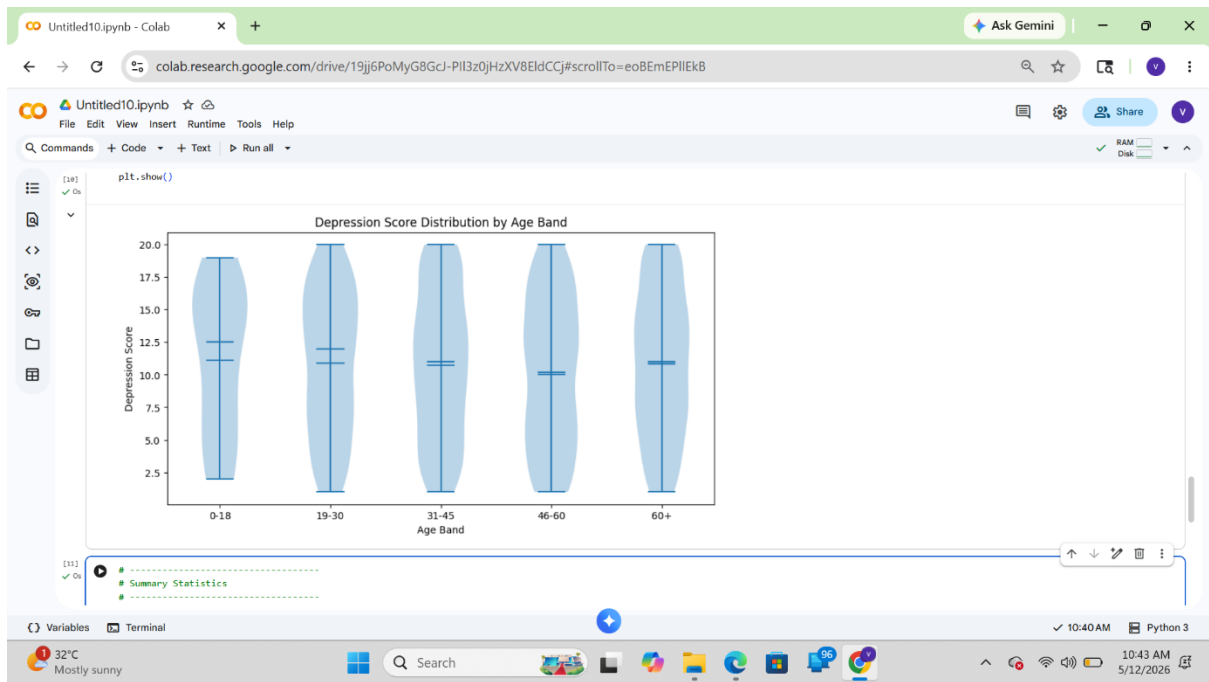
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```
[11]: # Summary Statistics
# Summary Statistics
# Summary Statistics

summary = df.groupby(
    ["Gender", "Age_Band"]
)[["Anxiety_Score", "Depression_Score"]].mean()

print(summary)
```

Variables Terminal

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Ask Gemini

File Edit View Insert Runtime Tools Help

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RAM Disk

```
[11]: )["Anxiety_Score", "Depression_Score"]].mean()
print(summary)
```

		Anxiety_Score	Depression_Score
Female	0-18	13.727273	11.181818
	19-30	11.006666	10.208696
	31-45	10.455172	11.006897
	46-60	10.812500	10.076389
	60+	10.532468	10.551948
Male	0-18	9.307692	11.000000
	19-30	10.413462	11.509615
	31-45	10.222973	10.662162
	46-60	9.393939	10.318182
	60+	10.813000	10.739837
Non-Binary	0-18	NaN	NaN
	19-30	11.750000	12.000000
	31-45	9.521739	10.217391
	46-60	10.500000	9.250000
	60+	11.074074	13.000000
Other	0-18	NaN	NaN
	19-30	10.000000	10.000000
	31-45	11.500000	8.666667
	46-60	9.666667	14.666667
	60+	8.000000	11.800000

/tmp/ipykernel_3966/3893271421.py:5: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current beh.

```
summary = df.groupby(
```

Variables Terminal

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1.Box Plot Analysis:

A box plot visually summarizes the distribution of PHQ/GAD scores using:

- **Median** → middle value
- **Quartiles (Q1 & Q3)** → spread of data
- **Whiskers** → range of normal values
- **Outliers** → unusually high or low scores
- The chart compares score distributions between males and females.
- If the female box is positioned higher, it indicates higher median anxiety/depression scores.
- Wider boxes indicate greater variability in mental health conditions.
- Outlier points represent individuals with extremely high PHQ/GAD scores.

2. Violin Plot Analysis:

A violin plot combines:

- Box plot statistics
- Probability density distribution

It shows both:

- central tendency
- distribution shape
- The violin plot shows how PHQ/GAD scores are distributed across age categories.
- Wider regions indicate many participants having similar scores.
- Narrow regions indicate fewer participants.
- Multiple peaks may suggest different mental health patterns within an age group.

Gender-Based Distribution Insights

Key Insights

- Compare median PHQ/GAD scores between genders
- Identify which group has higher symptom severity
- Observe variability in emotional distress
- Detect extreme mental health cases using outliers

Age-Based Distribution Insights

Key Insights

- Younger age groups often show broader distributions
- Middle-aged groups may show moderate median scores
- Older groups sometimes exhibit lower variability
- Helps identify vulnerable age populations

Conclusion

- **Box plots** are best for comparing medians, spread, and outliers.
- **Violin plots** provide deeper insight into data density and score distribution shape.
- Combining both charts gives a clearer understanding of PHQ/GAD patterns across gender and age demographics.
- These analyses help researchers and clinicians identify trends in anxiety and depression prevalence within populations.